Exercises 2.1

1. $e^{ix}z \in \mathbb{C}$ for all $z \in \mathbb{C}$.

   The identity in $(\mathbb{R}, +)$ is 0, and $0(z) = e^{i0}z = z$ for all $z \in \mathbb{C}$.

   $x(y(z)) = e^{ix}y(z) = e^{ix}e^{iy}z = e^{i(x+y)}z = (x+y)(z)$ for all $x, y \in \mathbb{R}$, $z \in \mathbb{C}$.

   Hence we have a group action.

   $\mathcal{O}(1) = \{x(1) \;:\; x \in \mathbb{R}\} = \{e^{ix} \;:\; x \in \mathbb{R}\} = \{z \in \mathbb{C} \;:\; |z| = 1\}$ (the unit circle). $G_1 = \{x \in \mathbb{R} \;:\; x(1) = 1\} = \{x \in \mathbb{R} \;:\; e^{ix} = 1\} = \{2n\pi \;:\; n \in \mathbb{Z}\}$.

   $\mathrm{Fix}(0) = \{z \in \mathbb{C} \;:\; 0(z) = z\} = \{z \in \mathbb{C} \;:\; e^0 z = z\} = \mathbb{C}$.

2. From the vector space axioms we know that $av \in V$ for all real a, $1v = v$ and $a(bv) = (ab)v$. Thus $a(v) = av$ defines an action of $\mathbb{R}^*$ on V.

3. $n_1(n_2(x)) = n_1 n_2 x \neq (n_1 + n_2)(x)$, so we do not have a group action.

4. If $h(g) = gh$ then $h_1(h_2(g)) = h_1(gh_2) = gh_2h_1$ but $(h_1h_2)(g) = gh_1h_2$, so this is not a group action (unless H is abelian).

   If $h(g) = gh^{-1}$ then certainly $h(g) \in G$. $e(g) = ge^{-1} = ge = g$.

   $h_1(h_2(g)) = h_1(gh_2^{-1}) = gh_2^{-1}h_1^{-1} = g(h_1h_2)^{-1} = (h_1h_2)(g)$.

   Hence we have a group action.

5. The orbit of each element is $\{1, 2, 3\}$.

   $G_1 = \{\iota, (2 \quad 3)\}$, $G_2 = \{\iota, (1 \quad 3)\}$, $G_3 = \{\iota, (1 \quad 2)\}$.

   $\mathrm{Fix}(\iota) = \{1, 2, 3\}$, $\mathrm{Fix}((1 \quad 2 \quad 3)) = \mathrm{Fix}((1 \quad 3 \quad 2)) = \varnothing$, $\mathrm{Fix}((2 \quad 3)) = \{1\}$, $\mathrm{Fix}((1 \quad 2)) = \{3\}$, $\mathrm{Fix}((1 \quad 3)) = \{2\}$.

6. From basic Linear Algebra, when A is orthogonal we have $|A\mathbf{v}| = |\mathbf{v}|$. All matrices representing rotations about the origin and reflections in lines through the origin are orthogonal. The orbit of $\mathbf{v}$ is $\{\mathbf{u} \in \mathbb{R}^2 \;:\; |\mathbf{u}| = |\mathbf{v}|\}$. Geometrically, this is a circle with centre $\mathbf{0}$ and radius $|\mathbf{v}|$.

7. $k \in H$ and $k \in J$, so $k \in H \cap J$.

   $hk^{-1} \in H$ and $kj \in J$ so $(hk^{-1}, kj) \in H \times J$. $e(h, j) = (he, ej) = (h, j)$

   For any $k_1, k_2 \in H \cap J$, $k_1(k_2(h, j)) = k_1(hk_2{}^{-1}, k_2j) = ((hk_2{}^{-1})k_1{}^{-1}, k_1(k_2j)) = (h(k_1k_2)^{-1}, k_1k_2j) = (k_1k_2)(h, j)$.

   Thus the given rule defines an action of $H \cap J$ on $H \times J$.

8. (a) σ is a rotation through $\dfrac{2\pi}{3}$ about the centre.

   τ is a reflection in CL. The other symmetries are

   $\sigma^2 = (1 \quad 3 \quad 2)(4 \quad 6 \quad 5)$. Rotation through $\dfrac{2\pi}{3}$ in opposite sense to σ.

   $\sigma\tau = (1 \quad 2)(4 \quad 5)$. Reflection in BN.

   $\sigma^2\tau = (1 \quad 3)(4 \quad 6)$. Reflection in AM. $G \cong S_3$ (or D_3).

(b) $\mathcal{O}(1) = \mathcal{O}(2) = \mathcal{O}(3) = \{1, 2, 3\}$. $\quad \mathcal{O}(4) = \mathcal{O}(5) = \mathcal{O}(6) = \{4, 5, 6\}$
$G_1 = G_4 = \{\iota, \tau\}$, $G_2 = G_5 = \{\iota, \sigma^2\tau\}$, $G_3 = G_6 = \{\iota, \sigma\tau\}$.
$\mathrm{Fix}(\iota) = X$, $\mathrm{Fix}(\sigma) = \mathrm{Fix}(\sigma^2) = \varnothing$ (the empty set),
$\mathrm{Fix}(\tau) = \{1, 4\}$, $\mathrm{Fix}(\sigma\tau) = \{3, 6\}$, $\mathrm{Fix}(\sigma^2\tau) = \{2, 5\}$.

(c) $|\mathcal{O}(1)| = 3$, $|G_1| = 2$ and $3 \times 2 = 6 = |G|$.
The left cosets of G_1 are $G_1 = \{\iota, \tau\}$, $\sigma G_1 = \{\sigma, \sigma\tau\}$, $\sigma^2 G_1 = \{\sigma^2, \sigma^2\tau\}$.
Both elements of G_1 map 1 to 1; both elements of σG_1 map 1 to 2; both elements of $\sigma^2 G_1$ map 1 to 3.

(d) $\mathrm{Fix}(G) = \varnothing$ so $|\mathrm{Fix}(G)| = 0$. $\quad \mathcal{O}(1) = \{1, 2, 3\}$ and $\mathcal{O}(4) = \{4, 5, 6\}$.
$|G_1| = 2$ so $[G : G_1] = 3$. $|G_4| = 2$ so $[G : G_4] = 3$.
$|X| = 6 = 0 + 3 + 3$, verifying the orbit decomposition theorem.

9. $e(x) = x$, so $x \sim x$ for all $x \in X$. Thus $\sim$ is reflexive.

   Suppose $x_1 \sim x_2$, so $g(x_1) = x_2$ for some $g \in G$. Then $g^{-1} \in G$ as G is a group, and we have $g^{-1}(g(x_1)) = g^{-1}(x_2)$, so $(g^{-1}g)(x_1) = g^{-1}(x_2)$, so $e(x_1) = g^{-1}(x_2)$ and hence $g^{-1}(x_2) = x_1$. Thus $x_2 \sim x_1$, so $\sim$ is symmetric.

   Suppose $x_1 \sim x_2$ and $x_2 \sim x_3$, so $g_1(x_1) = x_2$ and $g_2(x_2) = x_3$ for some $g_1, g_2 \in G$. Then $g_2 g_1 \in G$ as G is a group, and we have $(g_2 g_1)(x_1) = g_2(g_1(x_1)) = g_2(x_2) = x_3$, so $x_1 \sim x_3$. Thus $\sim$ is transitive, so it is an equivalence relation on X.

   The equivalence class of $x \in X$ is $\{y \in X \ : \ x \sim y\} = \{y \in X \ : \ g(x) = y$ for some $g \in G\} = \mathcal{O}(x)$.

10. If x and y are in the same orbit then $\mathcal{O}(x) = \mathcal{O}(y)$. Thus by the orbit-stabiliser theorem, $|G_x| = \dfrac{|G|}{|\mathcal{O}(x)|} = \dfrac{|G|}{|\mathcal{O}(y)|} = |G_y|$. This does not imply $G_x = G_y$. For example, in Question 5, $\mathcal{O}(1) = \mathcal{O}(2) = \{1, 2, 3\}$ and $|G_1| = |G_2|$, but $G_1 \neq G_2$.

11. Let $x \in X$. If G is transitive on X then $y \in \mathcal{O}(x)$ for all $y \in X$, so $X \subset \mathcal{O}(x)$. By definition $\mathcal{O}(x) \subset X$, so $\mathcal{O}(x) = X$.

    Conversely, if $\mathcal{O}(x) = X$ for all $x \in X$ then all elements of X are in the same orbit, so for all $x, y \in X$ there exists $g \in G$ such that $g(x) = y$.

    If G is finite and acts transitively on G then, by the orbit-stabiliser theorem, for each $x \in G$ we have $|G|.|G_x| = |G|$, so $|G_x| = 1$ and thus G_x is a trivial group.

12. Certainly $gxg^{-1} \in G$ for all $g, x \in G$.

    $e(x) = exe^{-1} = x$ and $g_1(g_2(x)) = g_1 g_2(x) g_1^{-1} = g_1 g_2 x g_2^{-1} g_1^{-1} = (g_1 g_2) x (g_1 g_2)^{-1}$ $= (g_1 g_2)(x)$. Thus we have an action of G on G.

    $\widetilde{G}_G = \{g \in G \ : \ gxg^{-1} = x \text{ for all } x \in G\} = \{g \in G \ : \ gx = xg \text{ for all } x \in G\}$.

    $\mathrm{Fix}(G) = \{x \in G \ : \ gxg^{-1} = x \text{ for all } g \in G\} = \{x \in G \ : \ gx = xg \text{ for all } g \in G\}$. This is clearly the same set as $\widetilde{G}_G$.

13. The smallest prime factors of 21, 143, 175 are 3, 11, 5 respectively. Thus by Proposition 2.7, subgroups of index 3, 11, 5, i.e. of order 7, 13, 35 respectively, must be normal (assuming such subgroups exist).

14. Order 25: C_{25} and $C_5 \times C_5$. Order 49: C_{49} and $C_7 \times C_7$.

15. Let $S = \{s_1, \ldots, s_n\}$. Then $g(S) = \{gs_1, \ldots, gs_n\}$. If $gs_i = gs_j$ then $s_i = s_j$, so $g(S)$ has n distinct elements, hence $g(S) \in X$.

    $e(S) = eS = S$ and $g_1(g_2(S)) = g_1(g_2S) = (g_1g_2)S$ by the associativity of subset multiplication, so the rule gives an action of G on X.

    For this action, the stabiliser of a set S is $G_S = \{g \in G \ : \ gS = S\}$.

    Let $y \in G_Sx$. Then $y = gx$ for some $g \in G_S$, so if $x \in S$ then $y \in gS = S$.

    Then $x = g^{-1}y$, and $g^{-1} \in G_S$ as G_S is a group, so if $y \in S$ then $x \in S$.

    Thus if $x \in S$ then $y \in S$ for all $y \in G_Sx$ and so $G_Sx \subset S$, whilst if $x \notin S$ then $y \notin S$ for all $y \in G_Sx$ and so $G_Sx \cap S = \varnothing$. Hence either $G_Sx \subset S$ or $G_Sx \cap S = \varnothing$.

    It follows that every right coset of G_S is either contained in S or lies wholly outside S. We know that these cosets are disjoint and every element of G is in one of them, so S is the union of the ones which lie inside S.

    Each coset has order $|G_S|$, so this divides $|S|$.

16. By the orbit-stabiliser theorem the order of an orbit divides $|G|$, so as p is prime it can only be 1 or p.

    The orbits of order 1 together form $\mathrm{Fix}(G)$. The other orbits have order p. Suppose there are k of them.

    If p divides $|X|$, say $|X| = mp$, then by the orbit decomposition theorem $mp = |\mathrm{Fix}(G)| + kp$ so $|\mathrm{Fix}(G)| = (m-k)p$, hence p divides $|\mathrm{Fix}(G)|$.

17. (a) $a_1, \ldots, a_{p-1}$ can be chosen arbitrarily, with $|G|$ possible values for each, and then a_p must equal $a_{p-1}^{-1} \cdots a_1^{-1}$. Thus $|X| = |G|^{p-1}$.

    (b) Let $\mathbf{x} = (a_1, \ldots, a_p)$.
    The product of the entries in $g^k(\mathbf{x})$ is still e, so $g^k(\mathbf{x}) \in X$. $e\mathbf{x} = g^0\mathbf{x} = \mathbf{x}$ and $g^k(g^\ell\mathbf{x}) = g^{k+\ell}\mathbf{x} = (g^kg^\ell)\mathbf{x}$, so we have a group action.

    (c) $\mathbf{x}$ is fixed by all elements of C_p iff $\mathbf{x} = (x, x, \ldots, x)$ where $x^p = e$.
    Thus $|\mathrm{Fix}(C_p)|$ is the number of elements $x \in G$ such that $x^p = e$.
    By Question 16 $|\mathrm{Fix}(C_p)|$ is a multiple of p, so there is more than one element $x \in G$ such that $x^p = e$. One such element is e, the only one whose order is 1. The other(s) have order p.
    The number of these is $kp - 1$ for some $k \in \mathbb{N}$, so it is congruent to -1 (mod p).

    (d) By (c) a group G of order $99 = 11 \times 9$ has at least one element of order 11. Such an element generates a cyclic subgroup H of order 11, and index 9, in G. Thus by Proposition 2.5 there is a homomorphism ϕ from G into S_9 such that $\mathrm{Ker}\,\phi$ is a subgroup of H. As $|H|$ is prime, $\mathrm{Ker}\,\phi$ is either trivial or all of H.
    Now if $\mathrm{Ker}\,\phi$ is trivial, ϕ is injective so $\phi(G) \cong G$. But 99 does not divide 9! so S_9 cannot have a subgroup isomorphic to G. Hence $\mathrm{Ker}\,\phi = H$, so $H \lhd G$.